

HR Employee Attrition Analysis & Prediction Report

IBM HR Analytics Dataset | 1,470 Employees

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1. Project Background

Employee attrition is one of the most costly challenges facing organizations today. Replacing a single employee costs between 6 and 9 months of their salary in recruitment, onboarding, and lost productivity. This project analyses IBM's HR dataset of 1,470 employees to identify the root causes of attrition, predict which employees are at risk of leaving, and provide actionable recommendations for HR leadership.

2. Dataset Overview

Field	Details
Source	IBM HR Analytics (Kaggle)
Total Employees	1,470
Total Features	35
Target Variable	Attrition (Yes/No)
Employees Who Left	237 (16.1%)
Employees Who Stayed	1,233 (83.9%)

3. Excel Analysis — Pivot Table Findings

Three pivot tables were created to explore attrition patterns across departments, job roles, and overtime status.

Attrition by Department

Department	Stayed	Left	Attrition Rate
Human Resources	51	12	19.0%
Research & Development	828	133	13.8%
Sales	354	92	20.6%

Sales has the highest attrition rate at 20.6%, meaning 1 in 5 Sales employees leave the company.

Attrition by Job Role

Job Role	Attrition Rate
Sales Representative	39.8%
Human Resources	23.1%
Laboratory Technician	23.9%

Sales Executive	17.5%
Research Scientist	16.1%
Manager	4.9%
Research Director	2.5%

Sales Representatives have a crisis-level attrition rate of nearly 40%. Nearly 4 in 10 Sales Representatives leave the company.

Attrition by Overtime

Overtime	Stayed	Left	Attrition Rate
No	944	110	10.4%
Yes	298	127	30.5%

Employees working overtime are 3 times more likely to quit than those who do not.

4. Python Analysis — Key Findings

Python was used in Google Colab for deeper exploratory data analysis and correlation analysis using Pandas and Matplotlib.

Attrition Rate by Department

The bar chart confirmed the Sales department has the highest attrition rate at approximately 20%, followed by Human Resources at 19% and Research & Development at 13.8%.

Attrition Rate by Overtime

The overtime analysis confirmed that 30.5% of overtime workers leave compared to only 10.4% of non-overtime workers. This is the single most actionable finding in the entire dataset because it is a direct HR policy problem the company can fix immediately.

Age Distribution Analysis

Younger employees aged 25 to 35 show the highest rate of attrition. This is consistent with early-career job-hopping behaviour, in which employees explore multiple opportunities before settling.

Monthly Income Analysis

Employees who left consistently earned lower monthly income across all age groups. The box plot showed that the median income of employees who stayed was significantly higher than that of those who left, confirming that compensation is a key retention factor.

Marital Status

Single employees leave at 25.3% compared to 12.4% for married and 10.1% for divorced employees. Single employees have fewer financial commitments and are more willing to take risks on new opportunities.

Correlation Analysis

The correlation chart identified the following key relationships with attrition:

Factors that increase attrition risk:

- Distance from home (positive correlation)
- Number of companies worked (positive correlation)

Factors that reduce attrition risk:

- Total working years (strong negative correlation)
- Job level (negative correlation)
- Years in current role (negative correlation)
- Monthly income (negative correlation)
- Age (negative correlation)

5. SPSS Modeler — Predictive Model

A C5.0 Decision Tree classification model was built in IBM SPSS Modeler 18.0 to predict which employees are likely to leave.

Model Configuration

Setting	Value
Algorithm	C5.0 Decision Tree
Output Type	Classification
Tree Depth	9 levels
Target Variable	Attrition
Input Variables	31 employee features

Model Accuracy

Result	Count	Percentage
Correct Predictions	1,350	91.84%
Incorrect Predictions	120	8.16%
Total	1,470	100%

The model achieved 91.84% accuracy on the training data.

Top Predictors of Attrition (by importance)

1. TotalWorkingYears — most important
2. MaritalStatus
3. JobLevel
4. JobSatisfaction

5. YearsSinceLastPromotion
6. Age
7. MonthlyRate
8. NumCompaniesWorked
9. DailyRate
10. WorkLifeBalance — least important among the top 10

The first decision split in the tree is at TotalWorkingYears ≤ 2 , meaning employees with 2 or fewer years of total work experience are the highest risk group for attrition.

6. Tableau Dashboard

An interactive dashboard was built in Tableau Public with 5 visualizations and cross-chart filtering:

- Attrition by Department (stacked bar, percentage view)
- Attrition by Job Role (horizontal bar, sorted by rate)
- Overtime Impact (comparative bar chart)
- Age vs Monthly Income (scatter plot, colored by attrition)
- Attrition by Marital Status (stacked bar)

Live Dashboard:

https://public.tableau.com/app/profile/muskan.choudhary8375/viz/HRAAttritionAnalysisDashboard_17825895443370/HRAAttritionAnalysisDashboard

7. Business Recommendations

1. Address Overtime Immediately

Overtime workers leave at 3x the rate. Audit which teams are consistently working overtime and either hire additional staff or restructure workloads. This is the fastest win available.

2. Sales Representative Retention Program

With 39.8% attrition, Sales Representatives are in crisis. Introduce clear career progression paths, competitive commission structures, and regular one-on-ones with managers to address concerns early.

3. Review Compensation for Junior Employees

The income analysis clearly shows that lower-paid employees leave more. Benchmark salaries against market rates, especially for Laboratory Technicians, Human Resources staff, and Sales Representatives.

4. Focus on New Employees

The SPSS model identified TotalWorkingYears ≤ 2 as the first split, meaning new employees are the highest risk group. Strengthen the onboarding program and assign mentors to all employees in their first two years.

5. Flexible Work for Distant Commuters

Distance from home is positively correlated with attrition. Introduce remote or hybrid work options for employees who live far from the office to reduce daily friction.

6. Promote from Within

YearsSinceLastPromotion is a predictor of attrition. Employees who feel stuck leave. Create transparent promotion criteria and fast-track high performers.

8. Conclusion

This analysis identified overtime, low compensation, sales role pressure, and lack of career progression as the primary drivers of attrition at IBM. The SPSS predictive model with 91.84% accuracy can be used by HR to proactively identify at-risk employees before they resign. Implementing the six recommendations above could meaningfully reduce the current 16.1% attrition rate toward the industry benchmark of 10-12%, saving the company significant recruitment and training costs.